

2(a)(i)	<i>any one from:</i> (faecal samples) can be, collected remotely / AW, without harming the chimpanzees / investigators ; (faecal samples) are, non-invasive / less stressful for chimpanzees / limits human interaction / no risk of infection / no need to anaesthetise chimpanzees ;	1
2(a)(ii)	<i>any two from:</i> make into a, suspension / solution / liquify faeces ; filtered / centrifuged (to remove solids) ;	2
2(a)(iii)	<i>idea of:</i> genetic profiling / DNA fingerprinting / DNA profiling / DNA sequencing / genetic analysis / (DNA) electrophoresis ;	1
2(b)	<i>any three from:</i> 1 (some) SIV antibodies will bind to HIV <u>antigens</u> ; 2 details of, which / number of lines / bands, match / don't match / AW ; 3 (some) SIV and HIV antibodies have similar (protein), structures / shape ; 4 HIV– people do not contain (HIV) antibodies / SIV– chimpanzees do not contain (SIV) antibodies ; 5 the tests are valid as there is a line in each of the control regions ;	3
2(c)(i)	1 allows for, valid / meaningful, comparison (between SIV+ and SIV– chimpanzees) ; 2 the groups have different, initial / original numbers ;	2
2(c)(ii)	no difference between the, life expectancy / AW, of chimpanzees that are SIV+ and SIV– ;	1

Question	Answer	Marks
2(c)(iii)	<i>mp1:</i> probability / it, is where null hypothesis is rejected ora ; <i>or</i> probability / it, is where critical values are chosen ; <i>mp2:</i> less than, 0.05 / 5% / 5 in 100, probability that the, result / it, is not significant / is due to chance ; <i>or</i> more than, 0.95 / 95% / 95 in 100, probability that the, result / it, is significant / is not due to chance ;	2
2(c)(iv)	<i>any two from:</i> <i>idea that:</i> 1 results, may not be due to death from SIV / may be due to disappearance ; 2 death could be due to, hunting / accident / fighting / predators / lack of food / disease (infectious or other) / AW ; 3 ages of chimps were estimated ; 4 uninfected chimpanzees may catch SIV during the investigation ; 5 no data on, male/female chimps (as life expectancy may differ) ;	2